# Spatio–Temporal–Spectral Collaborative Learning for Spatio–Temporal Fusion with Land Cover Changes

Xiangchao Meng, *Member, IEEE*, Qiang Liu, Feng Shao, *Member, IEEE*, and Shutao Li, *Fellow, IEEE*

***Abstract*— Spatio-temporal fusion by combining the complementary spatial and temporal advantages of multisource remote-sensing images to obtain time-series high spatial resolution images is highly desirable in monitoring surface dynamics. Currently, deep learning (DL)-based fusion methods have received extensive attention. However, existing DL-based spatio-temporal fusion methods are generally limited in fusing the images with land cover changes. In this article, we propose a spatio-temporal–spectral collaborative learning framework for spatio-temporal fusion to alleviate this problem. Specifically, the proposed method integrates the convolutional neural network and recurrent neural network into a unified framework, consisting of three subnetworks: the multiscale Siamese convolutional neural network (MSCNN), the multilayer convolutional recurrent neural network, and the adaptive weighting fusion network (AWFNet). The MSCNN has a flexible weight-sharing network to extract multiscale spatial–spectral features from multisource remote-sensing images. The multilayer convolutional recurrent neural network is constructed on the convolutional long short-term memory units to comprehensively learn the land cover changes by spatial, spectral, and temporal joint features. The AWFNet with a spatio-temporal–spectral change (STSC) loss is proposed to further improve the interpretability and robustness. The experiments were performed on the publicly available benchmark datasets featured phenology and land cover type changes, respectively. The experimental results demonstrated the competitive performance of the proposed method than other state-of-the-art fusion methods.**

***Index Terms*— Adaptive weighting fusion, convolutional neural network (CNN), land cover changes, recurrent neural network (RNN), spatio-temporal fusion.**

## I. Introduction

THE remote-sensing satellite images with high spatial and high temporal resolutions are highly desirable for monitoring surface dynamics and have a wide range of applications in monitoring land changes [1], [2], agricultural monitoring [3], [4], and urban change mapping [5], [6]. However, scarcely any sensors can provide images with both high spatial and high temporal resolutions due to technical limitations and other reasons [7], [8]. In addition, optical satellite images are inevitably affected by clouds and other atmospheric conditions, and this further hinders satellite sensors from acquiring dense time-series images with high precision. To solve the above-mentioned problems, researchers have put forward spatio-temporal fusion theory to combine the spatial advantage of high spatial resolution images (e.g., Landsat images) and the temporal advantage of low spatial resolution images (e.g., MODIS images) to generate high spatio-temporal resolution images. This has significantly improved the utilization of remote-sensing data for further mining of historical remote-sensing data and making full use of newly acquired images.

In recent years, spatio-temporal fusion technology has developed rapidly. Most spatio-temporal fusion methods can be broadly divided into the following four categories [9]: weight function-based methods, unmixing-based methods, Bayesian-based methods, and learning-based methods. For convenience, we refer to images with low spatial resolution as coarse images (e.g., MODIS images) and images with high spatial resolution as fine images (e.g., Landsat images).

The weight function-based methods combine the information of input images through a weight filtering function to estimate the target fine image. The spatial and temporal adaptive reflectance fusion model (STARFM) [10] introduces a weighting strategy by considering the neighboring pixels to generate the fine image, which is the first weight function-based method with the assumption that the change in reflectance is linear between the coarse and fine pixels. However, the STARFM generally achieves poor results in heterogeneous areas with diversified land cover changes. To solve the above problem, a new spatial and temporal adaptive algorithm for mapping reflectance change (STARCH) [11] was proposed, which requires at least two fine images to improve the accuracy of detecting reflectance changes. In the following research, the enhanced STARFM (ESTARFM) [12], the modified version of ESTARFM (mESTARFM) [13], and the integrating image inpainting and steering kernel regression fusion model (ISKRFM) [14] were proposed. Although many weight function-based methods have been proposed, the accuracy of prediction results is limited for the land cover changes in a complex scene.

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The unmixing-based methods reconstruct the fine image based on the linear spectral mixing theory. They generally consist of four steps: 1) definition of end members; 2) calculation of the abundance; 3) unmixing; and 4) reconstruction. The multisensor multiresolution technique (MMT) [15] is the first attempt based on the linear spectral unmixing theory. Afterward, many unmixing-based methods have been proposed for the MMT. For example, the spatio-temporal data fusion approach (STDFA) [16] predicted the target fine image by adding the land cover changes between two coarse images into the reference fine image, upon the assumption that the spatio-temporal change between coarse and fine images is constant. The enhanced spatio-temporal data fusion model (ESTDFM) [17] improved the robustness through the design of multiple weights with considering the impact of multiple temporal coarse images. Later, the unmixing-based data fusion (UBDF) model [18] and the improvement of UBDF [19] were proposed to improve the performance. On the whole, most of the unmixing-based methods have difficulties in achieving satisfactory results when the proportion of various surface features has distinct changes.

The Bayesian-based methods predict the target fine image based on the Bayesian estimation theory, which takes spatio-temporal fusion as a maximum *a posteriori* (MAP) problem. For example, Addesso *et al.* [20] introduced an interesting batch method for resolution enhancement, which combines deterministic interpolation and Bayesian estimation to obtain time-series thermal infrared images. Chu *et al.* [21] proposed a multidictionary Bayesian spatio-temporal reflectance fusion model (MDBFM) to improve the fusion performance in land cover changes by focusing on the edge information of the changed region. Li *et al.* [23] presented the spatio-temporal fusion model based on a sequential Bayesian estimation framework for thermal. In addition, there are also other Bayesian-based spatial and temporal relationship models, including covariance functions, low-pass filtering [24], linear mixing model [25], and joint covariance [26]. In addition, Shen *et al.* [27] proposed a variational-based unified framework for spatio-temporal-spectral fusion. On the whole, the Bayesian-based methods have a rigorous mathematical theory, but with relatively poor computational efficiency due to the iterative optimal solution.

The learning-based methods with encoding the features of coarse-fine image pairs to describe the relationship between the inputs and predicted images have emerged in recent years [28], [29]. The learning-based methods can be further subdivided into sparse representation (SR)- and deep learning (DL)-based methods. Early SR-based methods predict the target fine image on a pair of coupled dictionaries, such as the SR-based spatio-temporal reflectance fusion model (SPSTFM) [30] and the error-bound-regularized sparse coding (EBSPTM) [31]. Afterward, with the success of DL technology in the field of image classification and super-resolution, the DL-based methods have attracted ever-increasing attention. The spatio-temporal fusion method using a deep convolutional neural network (STFDCNN) [32] is the first attempt, which adopts a super-resolution convolutional neural network [33] to solve the large spatial resolution gaps between the coarse and fine images. Zhenyu *et al.* [34] proposed a deep convolutional spatio-temporal fusion network (DCSTFN). Then, the enhanced deep convolutional spatio-temporal fusion network (EDCSTFN) [35] was proposed to boost the prediction accuracy and enhance the fusion quality. Liu *et al.* [36] proposed a two-stream CNN (StfNet) by considering the temporal dependence and temporal consistency to predict the target images. Li *et al.* [37] integrated the dilated convolution and multiscale mechanisms into a network (DMNet) by considering the context information to improve the fusion accuracy. The spatio-temporal temperature fusion network (STTFN) [38] was proposed to enhance the potentially complex nonlinear relationship modeling between the inputs and predicted images by using a multiscale fusion CNN. However, all the above DL-based methods are trained separately for each band without considering the spectral correlations among bands. To tackle the problem and improve the accuracy, Chen *et al.* [39] proposed a multiscale two-stream CNN (STFMCNN) by integrating the multiscale CNN and temporal consistency into a unified framework, and they introduced a local fusion strategy to combine the two-stream predicted results. Besides CNN, several works on the generative adversarial network (GAN) [40] were proposed to feed all spectral bands into the trainable network. For instance, Zhang *et al.* [41] proposed a spatio-temporal fusion method based on GAN (STFGAN), which adopted a two-stage framework with super-resolution GAN (SRGAN) [42] for each stage. Tan *et al.* [43] proposed a GAN-based spatio-temporal fusion model (GAN-STFM) to reduce the number of model inputs and break through the time restriction on reference image selection. Chen *et al.* [44] proposed a spatio-temporal fusion model using Cycle-GAN (CycleGAN-STF) to improve the accuracy.

Although the DL-based methods have exhibited competitive performance, there are still several limitations. First, existing DL-based spatio-temporal fusion methods are insufficient in integrating complementary spatial, temporal, and spectral information. Correspondingly, most existing spatio-temporal fusion methods show limited performance for images with land cover changes. Second, most existing DL-based methods obtain the spatio-temporal change information of the coarse images through the pixel-level subtraction operation. However, the coarse spatio-temporal changes in this way cannot completely cover the fine spatio-temporal changes due to the general larger spatial resolution ratio ranging from 8 to 16, such as spatio-temporal fusion for Landsat and MODIS. Finally, existing DL-based spatio-temporal fusion methods [36]–[39] on five images generally first predict the fused images by a pair of fine-coarse images (before or after the target date) and the coarse image at the target date, respectively, and then the final fused image is obtained by designing a weighting function to fuse the two prediction results. This operation, on the one hand, cannot make full use of the complementary spatio-temporal advantages; on the other hand, it seems to be clumsy and rigid by requiring the training of two models.

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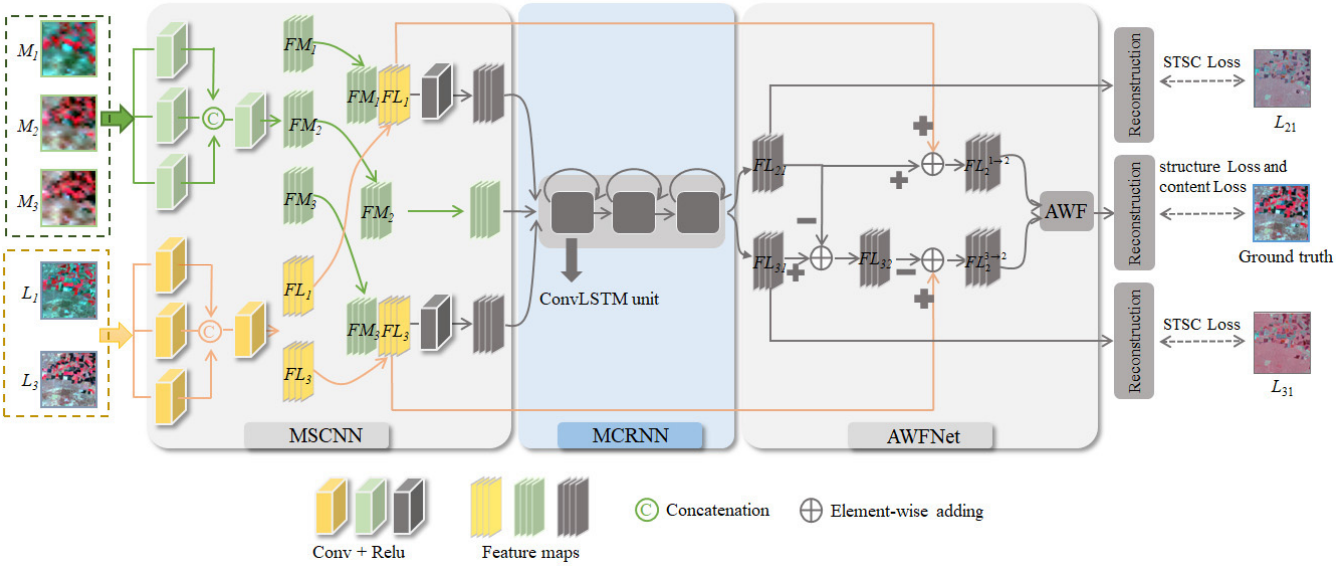



Fig. 1. Flowchart of the proposed STFRNN. ($M_i$ and $L_i$ are the MODIS and Landsat images at date $i$, respectively. $\mathrm{FM}_i$ and $\mathrm{FL}_i$ are the feature maps of $M_i$ and $L_i$, respectively. $\mathrm{FL}_{ij}$ denotes the feature maps of $L_{ij}$, which denotes the difference maps between $L_i$ and $L_j$ (e.g., $L_i - L_j$).)

In this article, we propose a spatio-temporal fusion method using a recurrent neural network (STFRNN) for land cover changes based on spatio-temporal–spectral collaborative learning. First, we design a multiscale Siamese CNN (MSCNN) to extract the multiscale spatial–spectral features for multisensor and multitemporal remote-sensing images. Second, we propose a multilayer convolutional recurrent neural network (MCRNN) to learn temporal changing information, which is constructed on convolutional long short-term memory (ConvLSTM) unit [45] to integrate the multitemporal spatial–spectral features. Third, we propose an adaptive weighting fusion network (AWFNet) to predict the high-precision fused images by jointly constraining the spatio-temporal–spectral change (STSC) features and the target fine image. To the best of our knowledge, it is the first attempt to employ RNN in the field of spatio-temporal fusion of remote-sensing images. Compared with the previous DL-based methods, the proposed STFRNN has the following advantages.

1) We propose a spatio-temporal fusion method for remote-sensing images with land cover changes based on spatio-temporal–spectral collaborative learning, which integrates the advantages of CNN and RNN into an end-to-end framework to simultaneously learn the spatio-temporal–spectral changing features from multisource and multitemporal. In the proposed STFRNN, a spatio-temporal–spectral collaborative learning framework is designed, and an adaptive weighting fusion module with STSC losses is proposed to enhance the interpretability and robustness. The experimental data in phenology and land cover type changes verified the performance.

The rest of the article is organized as follows. Section II presents the proposed method. The experimental results are presented and discussed in Section III, and conclusions are given in Section IV.

## II. Proposed Method

We propose a spatio-temporal–spectral collaborative learning framework for remote-sensing image spatio-temporal fusion with land cover changes. In the proposed method, five images including a coarse image at the target date and two pairs of fine-coarse images before and after the target date are used as the inputs. Taking the Landsat and MODIS images as fine and coarse images, respectively, the proposed STFRNN aims to reconstruct the Landsat image at the target date $t_2$ ($L_2$) with the corresponding MODIS image at date $t_2$ ($M_2$) and two Landsat-MODIS image pairs at neighboring date (e.g., $L_1$ and $M_1$, $L_3$ and $M_3$) as reference. As shown in Fig. 1, we first design a multiscale Siamese CNN (MSCNN) to extract the multiscale features of multitemporal and multisource fine and coarse remote-sensing images, respectively. And then integrate the multisource fine and coarse features at the same date to obtain the multitemporal spatial–spectral features. Then, we design the multilayer convolutional RNN (MCRNN) to effectively extract the changing features from the multitemporal spatial–spectral features. Finally, we propose an AWFNet to reconstruct the predicted image by integrating the temporal consistency strategy and an adaptive weighting fusion module.

### A. Multiscale Siamese CNN (MSCNN)

We integrate the multiscale mechanism and the Siamese network to design the MSCNN to extract the valuable features of five images (e.g., $M_2$, $L_1$ and $M_1$, $L_3$ and $M_3$). Considering that a ground scene in remote-sensing images may cover many objects of different sizes, we design a multiscale mechanism to improve the robustness of feature extraction in various ground scenes. Moreover, the Siamese network is employed to extract the features of multitemporal images captured from the same sensor, which can not only reduce the training parameters, but also highlight the valuable spatial–spectral features at three

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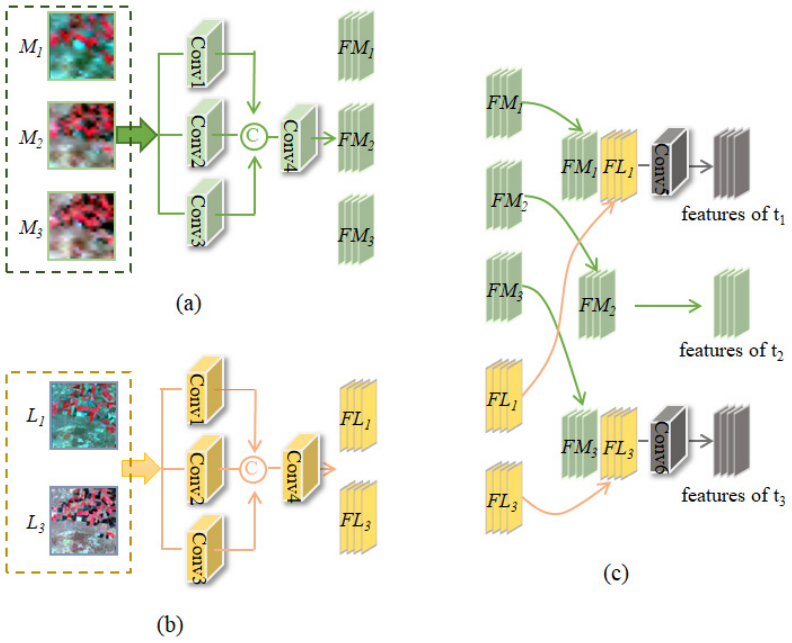



Fig. 2. Flowchart of the MSCNN. (a) Multiscale Siamese CNN to extract the features of MODIS images. (b) Multiscale Siamese CNN to extract the features of Landsat images. (c) Integrated spatial–spectral features at different phases.

time phases. In addition, the same architecture but without weight sharing network is employed to extract the features of images captured from different sensors, namely a pseudo-Siamese network. Specifically, a weight-sharing multiscale CNN is designed to extract the features of MODIS images (e.g., $M_1$, $M_2$, and $M_3$), as shown in Fig. 2(a), and the same architecture of weight-sharing multiscale CNN is employed to extract the features of Landsat images ($L_1$ and $L_3$), as shown in Fig. 2(b). It is worth mentioning that the weights of the network in Fig. 2(a) and (b) are not shared considering that the images are captured from different sensors.

The features of the five images (e.g., $FM_2$, $FL_1$ and $FM_1$, $FL_3$, and $FM_3$) are obtained through a multiscale Siamese feature extraction network. Then we integrate the features on the same date to obtain the spatial–spectral features at the three-time phases, as shown in Fig. 2(c). Table I shows the detailed parameters of convolutional layers in the spatial–spectral feature extraction module.

TABLE I
PARAMETERS OF CONVOLUTIONAL LAYERS IN THE MSCNN

| Name | Filter size | Filter number |
|---|---|---|
| Convolutional layer 1 | 3×3 | 32 |
| Convolutional layer 2 | 5×5 | 32 |
| Convolutional layer 3 | 7×7 | 32 |
| Convolutional layer 4 | 1×1 | 64 |
| Convolutional layer 5 | 3×3 | 64 |
| Convolutional layer 6 | 3×3 | 64 |

### *B. Multilayer Convolutional Recurrent Neural Network (MCRNN)*

In the previous methods [36], [38], the spatio-temporal changing information was obtained through pixel-level subtraction operation between the coarse images at different dates, which will reduce the robustness with abnormal or noise pixels. To solve the problem, we propose to learn the temporal changing information on the joint spatial, temporal, and spectral features of multitemporal images. The RNN can capture the dynamics of the sequence through the periodicity in the network, which is suitable for modeling the sequence with the temporal relationship. Recently, many variants of RNN have been proposed to solve the problem of long-term memory and vanishing gradients in backpropagation in RNN, such as the representative long short-term memory (LSTM) [46] and gate recurrent unit (GRU) [47]. However, the inputs of these models must be expanded into a 1-D vector, which will lose the spatial information of images. Considering this limitation, we introduce the convolutional LSTM for spatio-temporal modeling, which can collaboratively extract spatio-temporal–spectral features.

In this article, the MCRNN is established on the ConvLSTM unit, which is shown in Fig. 3. As can be seen, the ConvLSTM unit contains three gates (e.g., forget gate $f_t$, input gate $i_t$, and output gate $o_t$) and a cell state $C_t$, described as follows.

*Forget gate:* The ConvLSTM decides to discard the information from the cell state through the forget gate $f_t$, calculated by

$$f_t = \sigma(W_f * [H_{t-1}, X_t] + b_f) \tag{1}$$

where $\sigma$ denotes the sigmoidal function, [] represents the concatenate operation, and $*$ indicates the convolution operation. $H_{t-1}$ denotes the output of the hidden layer at date $t$, $X_t$ is

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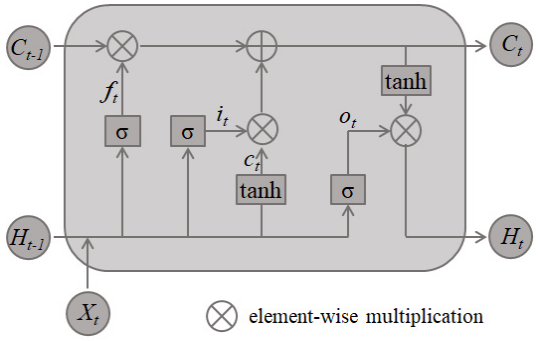



Fig. 3. Flowchart of the ConvLSTM.

the input of the hidden layer at date $t$, and $W_f$ and $b_f$ are the weights and bias of the current module, respectively.

*Input gate:* The input gate encodes the important information of the input data at the current date into the cell state, and which consists of two parts

$$i_t = \sigma(W_i * [H_{t-1}, X_t] + b_i) \tag{2}$$

$$c_t = \tanh(W_c * [H_{t-1}, X_t] + b_c) \tag{3}$$

where tanh denotes the tanh activation function, and $W$ and $b$ are the weights and bias of the current module, respectively.

*Cell state:* The information in the cell state can be updated through the input gate $i_t$ and the forget gate $f_t$, then $C_t$ is calculated by

$$C_t = f_t \otimes C_{t-1} + i_t \otimes c_t \tag{4}$$

where $C_t$ denotes the value of cell state of the hidden layer at date $t$, $\otimes$ denotes the element-wise multiplication, and $C_{t-1}$ denotes the value of cell state of the hidden layer at date $t-1$.

*Output gate:* The output gate $o_t$ determines the output value of the convLSTM unit, which consists of two parts

$$o_t = \sigma(W_o * [H_{t-1}, X_t] + b_o) \tag{5}$$

$$H_t = o_t \otimes \tanh(C_t) \tag{6}$$

where $H_t$ denotes the output of the hidden layer at date $t$, and $W_o$ and $b_o$ are the weights and bias of the current module, respectively.

As described above, the cell state $C_t$ is the core of the ConvLSTM, which saves changing information between multitemporal images in the spatio-temporal fusion task. In this article, we construct MCRNN using three-layer ConvLSTM to learn the changing features on multitemporal spatial and spectral features, as shown in Fig. 4. Especially, the outputs of the hidden layer at date 2 and date 3 (e.g., $H_2$ and $H_3$) are considered as the changing information (e.g., $\mathrm{FL}_{21}$, and $\mathrm{FL}_{31}$), respectively.

### C. Adaptive Weighting Fusion Network (AWFNet)

After the changing features (e.g., $\mathrm{FL}_{21}$, and $\mathrm{FL}_{31}$) are predicted, the predicted image is reconstructed based on the temporal consistency. Although the temporal consistency is applied in the pixel domain [36], we believe that it is also feasible in the feature domain and has higher robustness. The

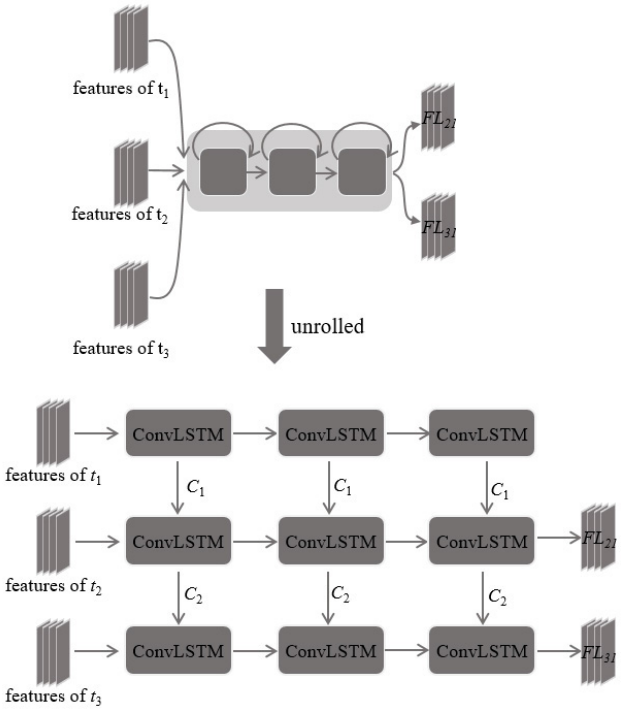



Fig. 4. Flowchart of the MCRNN.

feature differences of Landsat images on different dates can be calculated by

$$\mathrm{FL}_{21} = \mathrm{FL}_2 - \mathrm{FL}_1 \tag{7}$$

$$\mathrm{FL}_{31} = \mathrm{FL}_3 - \mathrm{FL}_1 \tag{8}$$

$$\mathrm{FL}_{32} = \mathrm{FL}_3 - \mathrm{FL}_2. \tag{9}$$

The two features of predicted images (e.g., $\mathrm{FL}_2^{1\to 2}$ and $\mathrm{FL}_2^{3\to 2}$) can be calculated by

$$\mathrm{FL}_{32} = \mathrm{FL}_{31} - \mathrm{FL}_{21} \tag{10}$$

$$\mathrm{FL}_2^{1\to 2} = \mathrm{FL}_1 + \mathrm{FL}_{21} \tag{11}$$

$$\mathrm{FL}_2^{3\to 2} = \mathrm{FL}_3 - \mathrm{FL}_{32} \tag{12}$$

where $\mathrm{FL}_2^{1\to 2}$ is the features of the predicted image at the target date with the help of the image pairs at the date $t_1$, and $\mathrm{FL}_2^{3\to 2}$ is the features of the predicted image at the target date with the help of the image pairs at the date $t_3$.

In other DL-based spatio-temporal fusion methods that take five images as inputs, the common operation is to design a weighting function based on the correlation of the MODIS images to fuse the two prediction results before and after the target date. In this article, we propose an adaptive weighting fusion module, in which the weights are adaptively learned to adjust the contributions of two prediction features, as shown in Fig. 5. In particular, we first concatenate the two features $\mathrm{FL}_2^{3\to 2}$ and $\mathrm{FL}_2^{1\to 2}$ as $\mathrm{FL}_2$, and then the global average pooling (GAP) and the global variance pooling (GVP) are employed to estimate the weights, in which the global mean and the variance of each branch are used for the estimation.

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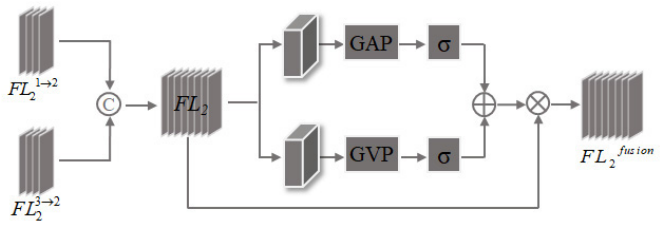



Fig. 5. Illustration of the adaptive weighting fusion module.

TABLE II
PARAMETERS OF CONVOLUTIONAL LAYERS IN THE AWFNET

| Name | Filter size | Filter number |
|---|---|---|
| Convolutional layer 1 | 3×3 | 64 |
| Convolutional layer 2 | 3×3 | 32 |
| Convolutional layer 3 | 3×3 | bands |

The calculations are illustrated as follows:

$$\mathbf{w}_{\text{avg}} = \sigma(\text{GAP}(f^{1\times1}(\text{FL}_2))) \tag{13}$$

$$\mathbf{w}_{\text{var}} = \sigma(\text{GVP}(f^{1\times1}(\text{FL}_2))) \tag{14}$$

$$\text{FL}_2^{\text{fusion}} = (\mathbf{w}_{\text{avg}} + \mathbf{w}_{\text{var}}) \otimes \text{FL}_2 \tag{15}$$

where GAP denotes the GAP, $f^{1\times1}$ denotes the convolution operation with the convolution kernel size of $1 \times 1$, GVP indicates the GVP, and $\otimes$ represents the element-wise multiplication.

After the fusion features $\text{FL}_2^{\text{fusion}}$ obtained through the adaptive weighting fusion module, the target image is reconstructed on three convolutional layers. The detailed parameters are listed in Table II, where the bands denote the channel dimension size of the target image.

### D. Loss Function

In this article, the overall loss function of the proposed STFRNN is composed of structure loss, content loss, and STSC loss, as shown in (16). The structure loss and content loss are employed to guarantee the consistency in structure and content between the predicted image and ground truth, respectively. The structure loss and content loss are calculated according to (17) and (18), respectively:

$$\mathcal{L}_{\text{total}} = \alpha\mathcal{L}_{\text{content}} + \beta\mathcal{L}_{\text{structure}} + \lambda\mathcal{L}_{\text{stsc}} \tag{16}$$

$$\mathcal{L}_{\text{content}} = \frac{1}{B}\sum_{b=1}^{B}\left|L_2^b - F_2^b\right| \tag{17}$$

$$\mathcal{L}_{\text{structure}} = 1 - \text{SSIM}(L_2, F_2) \tag{18}$$

where $b$ denotes the $b$th band of the image, B is the total number of bands, SSIM denotes the structural similarity index measure, $L_2$ denotes the observed Landsat image at the target date $t_2$, and $F_2$ denotes the predicted Landsat image at $t_2$. $\alpha$, $\beta$, and $\lambda$ are the weight coefficients, and we set $\alpha = \beta = \lambda = 1$ in the experiment.

To guarantee that the two features (e.g., $\text{FL}_{21}$, and $\text{FL}_{31}$) predicted by the MCRNN contain more accurate fine changing information, we utilize $\text{FL}_{21}$ and $\text{FL}_{31}$ to reconstruct the difference images $\text{PL}_{21}$ and $\text{PL}_{31}$, respectively. Then we calculate the MSE of the reconstructed difference images (e.g., $\text{PL}_{21}$, and $\text{PL}_{31}$) and the real difference images (e.g., $L_{21}$, and $L_{31}$) as the STSC loss, as illustrated in (19)–(21). The two reconstruction networks are composed of two convolutional layers, and the parameter of the convolutional layer is shown in Table III:

TABLE III
PARAMETERS OF CONVOLUTIONAL LAYERS IN THE RECONSTRUCTION NETWORK

| Name | Filter size | Filter number |
|---|---|---|
| Convolutional layer 1 | 3×3 | 32 |
| Convolutional layer 2 | 3×3 | band |

$$\mathcal{L}_{\text{stsc}}^{21} = \frac{1}{B}\sum_{i=1}^{b}\left[L_{21}^b - \text{PL}_{21}^b\right]^2 \tag{19}$$

$$\mathcal{L}_{\text{stsc}}^{31} = \frac{1}{B}\sum_{i=1}^{b}\left[L_{31}^b - \text{PL}_{31}^b\right]^2 \tag{20}$$

$$\mathcal{L}_{\text{stsc}} = \mathcal{L}_{\text{stsc}}^{21} + \mathcal{L}_{\text{stsc}}^{31} \tag{21}$$

where $\text{PL}_{31}$ and $\text{PL}_{21}$ denote the reconstructed difference images, respectively, and $L_{21}$ and $L_{31}$ are the real difference images, respectively.

## III. EXPERIMENTS

To comprehensively verify the proposed fusion model, the datasets featured by changes in phenology and land cover types, respectively, were employed in the experiments. Besides, the proposed fusion framework with RNN and GRU structures was analyzed to further verify the proposed model with the ConvLSTM. In addition, the discussions are given to further discuss the unique parts and the advantages and disadvantages of the proposed STFRNN.

### A. Datasets and Training Details

We conducted experiments on two publicly available benchmark datasets [48]. In the two datasets, the MODIS images have been processed using the nearest neighbor algorithm to match the Landsat data resolution. The first dataset is named as coleambally irrigation area (CIA) with rich change in phenological, but with fewer land cover changes. The CIA dataset consists of 17 MODIS-Landsat image pairs, which were captured from 2001 to 2002 with a dimension of size $1720 \times 2040 \times 6$. The whole dataset is also divided into two parts: training and testing sets. The training set consists of images at the first 13 dates, and the testing set consists of images at the last four dates.

The second dataset is named as lower Gwydir catchment (LGC) with significant regular shape changes in land cover due to the occurrence of a large flood in mid-December 2004. The LGC dataset consists of 14 MODIS-Landsat image pairs, which were captured from 2004 to 2005 with a spatial

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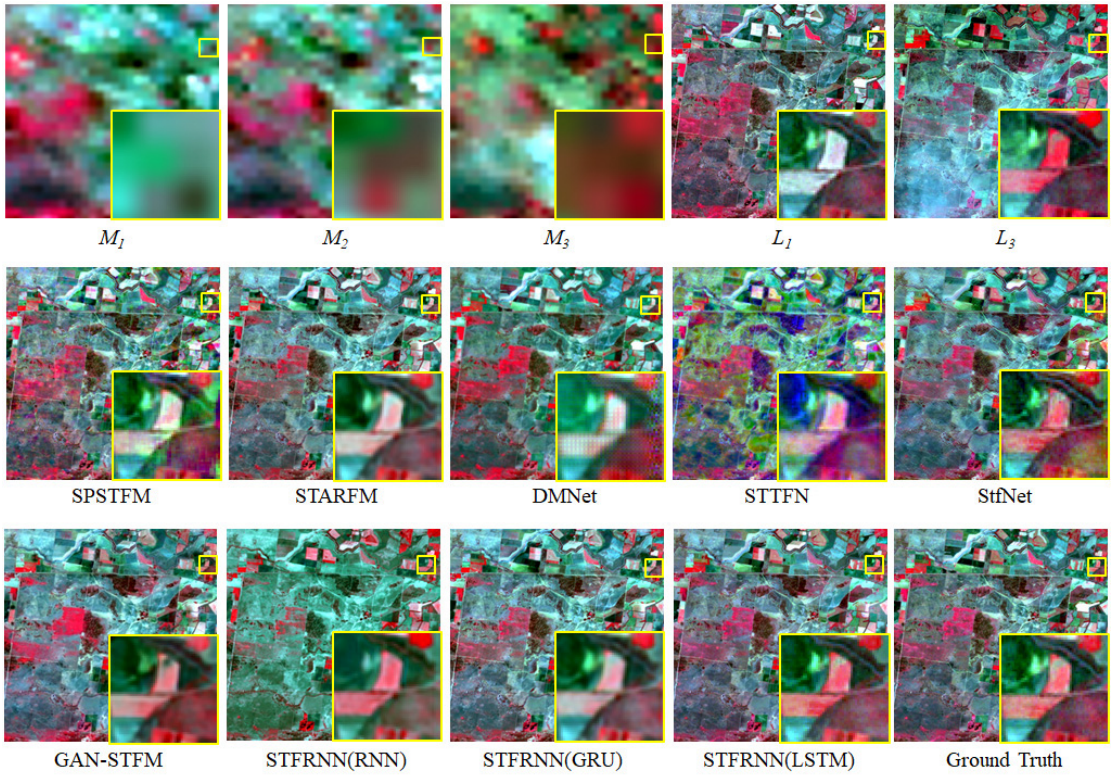



Fig. 6. Fusion results on April 18, 2002, in the CIA dataset. ($M_1$, $M_2$, and $M_3$ are the MODIS images captured on April 11, 2002, April 18, 2002, and April 27, 2002, respectively. $L_1$ and $L_3$ are the Landsat images captured on April 11, 2002, and April 27, 2002, respectively.)

dimension of 3200 × 2720 and a spectral dimension of six bands. The whole dataset is divided into two parts: a training set and a testing set. The training set consists of images at the first ten dates, and the testing set consists of images at the last four dates.

For the network training of the proposed STFRNN, the training set was cropped into small patches with a dimension size of 64 × 64 × 6 to minimize the memory consumption of the network, and the testing set was cropped into the patches with dimension size of 512 × 512 × 6 for comparison. During the training phase, the epoch was set to 200, the batch size was set to 8, and the initial learning rate was 0.001 with exponential decay. For a fair comparison, all the compared DL-based methods (e.g., DMNet, StfNet, STTFN, GAN-STFM, and the proposed STFRNN) were implemented in PyTorch 1.6.0 and trained on a GPU NVIDIA P5000 with 16-GB RAM. The STARFM and SPSTFM in the experiments were performed in the MATLAB R2017b framework on a PC with Intel Xeon Gold 5218 and 64-GB RAM.

### *B. Compared Methods and Evaluation Indices*

To illustrate the advantages of the proposed STFRNN, several state-of-the-art spatio-temporal fusion models, including two typical methods (e.g., STARFM [10], SPSTFM [30]) and four DL-based methods (e.g., DMNet [37], StfNet [36], STTFN [38], GAN-STFM [43]) were selected for comparison. In addition, two variants of the proposed STFRNN were also employed by replacing the ConvLSTM with the typical convolutional RNN and convolutional GRU, termed STFRNN (RNN) and STFRNN (GRU), respectively. All the experimental comparisons were conducted on the same testing set and the default parameters were employed.

For quantitative evaluation, several widely used evaluation indicators were employed to measure the quality of spatio-temporal fusion models. The root-mean-square error (RMSE) measures the fusion errors between the predicted image and the ground truth, in which a small value of RMSE indicates a better accuracy. The spectral angle mapper (SAM) [49] measures the spectral similarity, and a small value of SAM indicates a higher spectral fidelity. The structure similarity (SSIM) [50] measures the structural similarity, and the larger value of the SSIM, the similar spatial structure with the ground truth.

### *C. Experimental Results and Analysis*

The CIA dataset featured changes in phenology and the LGC dataset with significant regular shape changes in land cover were employed in the experiments, to evaluate the performance of the proposed STFRNN method.

*1) Experiments for Phenological Changes:* We conducted experiments on the CIA dataset to evaluate the performance of STFRNN with the changes in phenology. All the spatio-temporal fusion models aim to predict the target images on April 18, 2002, and April 27, 2002, respectively. The inputs and different fusion results are illustrated in Figs. 6 and 7. We can observe that all the methods can effectively predict

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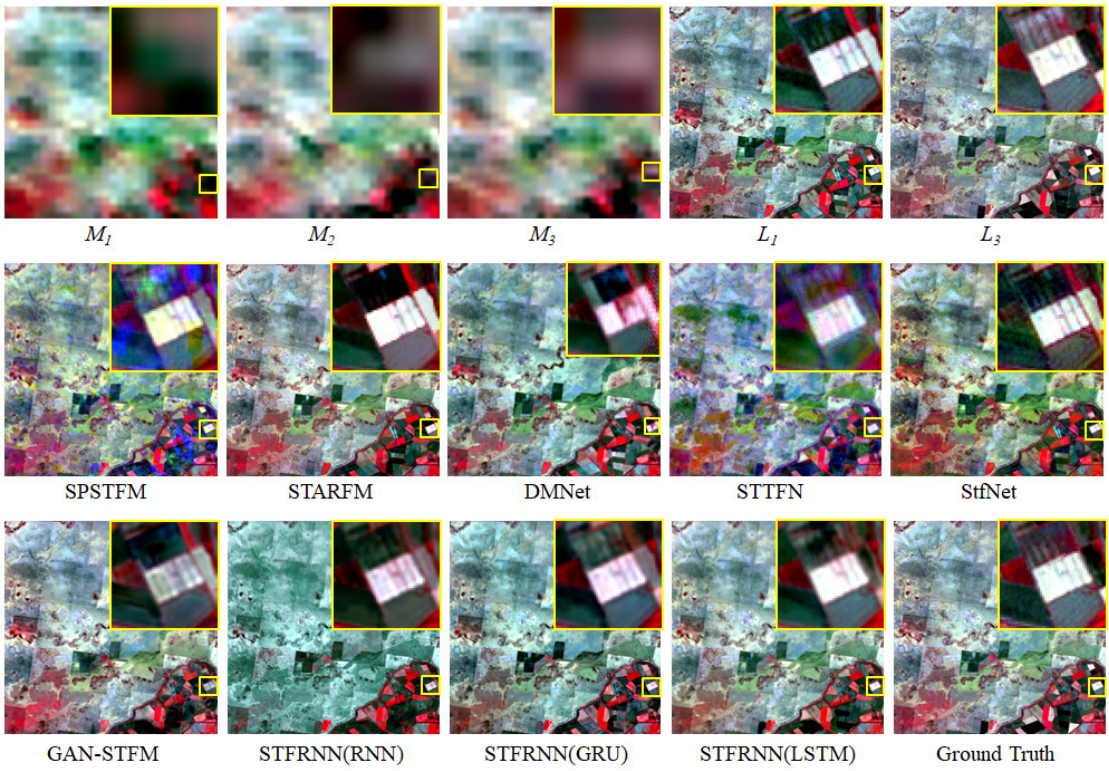



Fig. 7. Fusion results on April 27, 2002, in the CIA dataset. ($M_1$, $M_2$, and $M_3$ are the MODIS images captured on April 18, 2002, April 27, 2002, and May 4, 2002, respectively. $L_1$ and $L_3$ are the Landsat images captured on April 18, 2002, and May 4, 2002, respectively.)

the target image, but there are different performances in some areas. Specifically, it can be seen from Fig. 6 that the SPSTFM and STARFM show lower accuracy in predicting phenological changes in some areas (shown by the zoomed area). The DMNet has inconsistent errors and block effects. The STTFN achieves the worst result due to the assumption that the training parameters of the image at the neighboring date are also applicable to the current date. Although the result of StfNet is very similar to the ground truth, there is still spectral distortion in the predicted image. The GAN-STFM has a better performance in terms of spectral and spatial preservation, but it also cannot accurately predict phenological changes in some areas. The proposed STFRNN with LSTM achieves the best performance in spectral and spatial preservation, accurately predicting phenological changes. By contrast, the proposed framework constructed with RNN and GRU [e.g., STFRNN (RNN), STFRNN (GRU)] is slightly poor than the proposed STFRNN. Similarly, the same conclusion can be drawn from Fig. 7 that the proposed STFRNN still achieves the best performance in predicting the changes in phenology. To illustrate the advantages of the proposed STFRNN more clearly, we listed the heat map of SAM in Figs 8 and 9. As seen, the proposed STFRNN can achieve the best performance with the least spectral error.

In addition to qualitative experiments, quantitative experiments also play an important role in verifying the effectiveness of the fusion methods. The results of the quantitative evaluation for the CIA dataset are listed in Table IV. The proposed STFRNN achieves the lowest RMSE, SAM, and the highest SSIM values, which means that the proposed STFRNN provides excellent performance in terms of content, spectrum, and structure. In addition, the quantitative results obtained by the proposed framework with two variants [i.e., the STFRNN (RNN) and STFRNN (GRU)] also have competitive advantages compared with other methods. This illustrates that the spatio-temporal change information through learning instead of subtraction operation can obtain more accurate prediction results in predicting phenological changes.

*2) Experiments for Land Cover Type Changes:* The LGC dataset characterized by changes in land cover was adopted to evaluate the performance of the proposed STFRNN. All the spatio-temporal fusion models aim to predict the target images on February 14, 2005, and March 2, 2005, respectively. The inputs and different fusion results are illustrated in Figs. 10 and 11. We can see from the images of $L_1$ and $L_3$ that there are land cover changes during this period. Specifically, as can be seen from Fig. 10, the result of SPSTFM has spectral distortion and errors in some areas, which may be the reason that the SR-based method is not accurate enough in modeling complex land cover changes. The result of STARFM maintained well spectral and spatial information, but cannot predict the changes in land cover accurately. The result of DMNet suffers from spectral distortion and blurring in many areas. The STTFN and StfNet have spectral distortion and cannot predict the changes accurately due to the shallow network and the weighting strategy designed in the manual. The result of GAN-STFM suffers from spectral distortion severely due to insufficient spatio-temporal change information learned only from two images are used as inputs. The proposed STFRNN and its two variants of STFRNN (RNN) and STFRNN (GRU)

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TABLE IV

QUANTITATIVE EVALUATION RESULTS IN THE CIA DATASET ON APRIL 18, 2002 (THE BEST IS MARKED IN BOLD, AND THE SECOND BEST IS MARKED IN UNDERLINE)

| | Band | SPSTFM | STARFM | DMNet | StfNet | STTFN | GAN-STFM | STFRNN (RNN) | STFRNN (GRU) | STFRNN (LSTM) |
|---|---|---|---|---|---|---|---|---|---|---|
| RMSE | Band 1 | 0.0132 | 0.0180 | 0.0150 | 0.0142 | 0.0169 | 0.0107 | 0.0108 | 0.0099 | **0.0088** |
| | Band 2 | 0.0156 | 0.0199 | 0.0246 | 0.0188 | 0.0290 | 0.0138 | 0.0135 | 0.0130 | **0.0115** |
| | Band 3 | 0.0215 | 0.0256 | 0.0298 | 0.0251 | 0.0198 | 0.0204 | 0.0215 | 0.0199 | **0.0172** |
| | Band 4 | 0.0341 | 0.0386 | 0.0322 | 0.0350 | 0.0312 | 0.0332 | 0.0341 | 0.0295 | **0.0278** |
| | Band 5 | 0.0428 | 0.0451 | 0.0416 | 0.0392 | 0.0346 | 0.0408 | 0.0408 | 0.0354 | **0.0334** |
| | Band 6 | 0.0355 | 0.0410 | 0.0324 | 0.0362 | 0.0306 | 0.0371 | 0.0327 | 0.0312 | **0.0296** |
| | Average | 0.0271 | 0.0314 | 0.0293 | 0.0281 | 0.0270 | 0.0260 | 0.0256 | 0.0232 | **0.0214** |
| SSIM | Band 1 | 0.9229 | 0.9259 | 0.9079 | 0.9265 | 0.8722 | 0.9319 | 0.9346 | 0.9413 | **0.9482** |
| | Band 2 | 0.9149 | 0.9204 | 0.8709 | 0.9114 | 0.8512 | 0.9199 | 0.9263 | 0.9317 | **0.9359** |
| | Band 3 | 0.8644 | 0.8721 | 0.8433 | 0.8592 | 0.8727 | 0.8622 | 0.8759 | 0.8839 | **0.8928** |
| | Band 4 | 0.8005 | 0.8101 | 0.8384 | 0.8080 | 0.8124 | 0.8039 | 0.8240 | 0.8396 | **0.8450** |
| | Band 5 | 0.7384 | 0.7587 | 0.7769 | 0.7674 | 0.7785 | 0.7344 | 0.7582 | 0.7832 | **0.7885** |
| | Band 6 | 0.7520 | 0.7573 | 0.7585 | 0.7608 | 0.7825 | 0.7456 | 0.7818 | 0.7899 | **0.7966** |
| | Average | 0.8322 | 0.8407 | 0.8327 | 0.8389 | 0.8282 | 0.8330 | 0.8501 | 0.8616 | **0.8679** |
| SAM | All | 0.1757 | 0.1776 | 0.1764 | 0.1773 | 0.1982 | 0.1667 | 0.1670 | 0.1534 | **0.1419** |

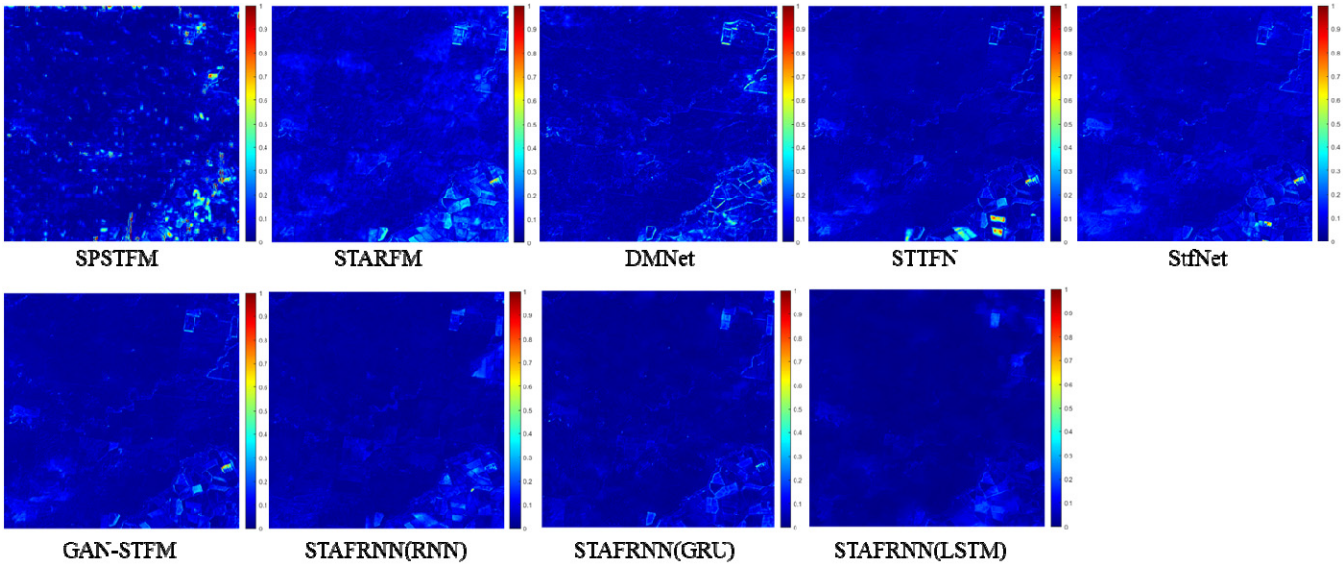



Fig. 8. Heat map of SAM on April 18, 2002, in the CIA dataset.

all achieve more similar results with the ground truth. Among the three results, the proposed STFRNN constructed with ConvLSTM obtains the best results. Similarly, the same conclusion can be found in Fig. 11 that the proposed STFRNN still achieves the most similar result to the ground truth, which illustrates that the proposed STFRNN has high robustness in predicting the changes in land cover. To illustrate the advantages of the proposed STFRNN more clearly, we listed the heat map of SAM in Figs 12 and 13. As seen, the proposed STFRNN can achieve the best performance with the least spectral error.

The quantitative evaluation results on February 14, 2005, and March 2, 2005, are listed in Table V. Similar to the conclusion in the CIA dataset, the proposed STFRNN achieves the best scores in all indicators. We can also find that the quantitative results obtained by the methods constructed with RNN [i.e., STFRNN (RNN), STFRNN (GRU)] have great advantages. This phenomenon illustrates that the spatio-temporal relationship model established by RNN has higher robustness and accuracy in detecting land cover changes.

In conclusion, the proposed STFRNN achieves the best performance compared with other state-of-the-art spatio-temporal fusion methods (i.e., STARFM, SPSTFM, DMNet, STTFN, StfNet, and GAN-STFM). Moreover, the proposed STFRNN achieves the best performance in predicting changes in both phenological and land cover.

### *D. Discussion*

*1) Ablation Study:* In the proposed STFRNN, an adaptive weighting fusion module was proposed. To validate the validity of the AWF module, we replace the AWF structure with concatenating operation, termed $\text{STFRNN}_{\text{noAWF}}$. In addition, we design an STSC loss function to improve the accuracy. To validate the validity of the proposed STSC loss function,

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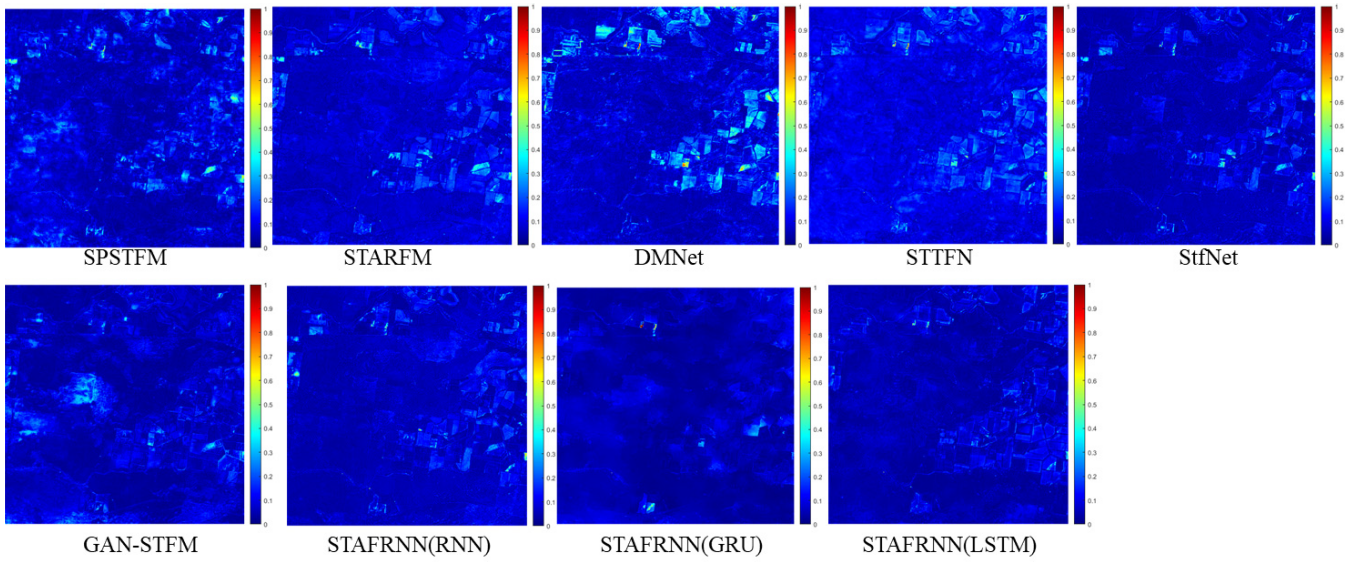



Fig. 9. Heat map of SAM on April 27, 2002, in the CIA dataset.

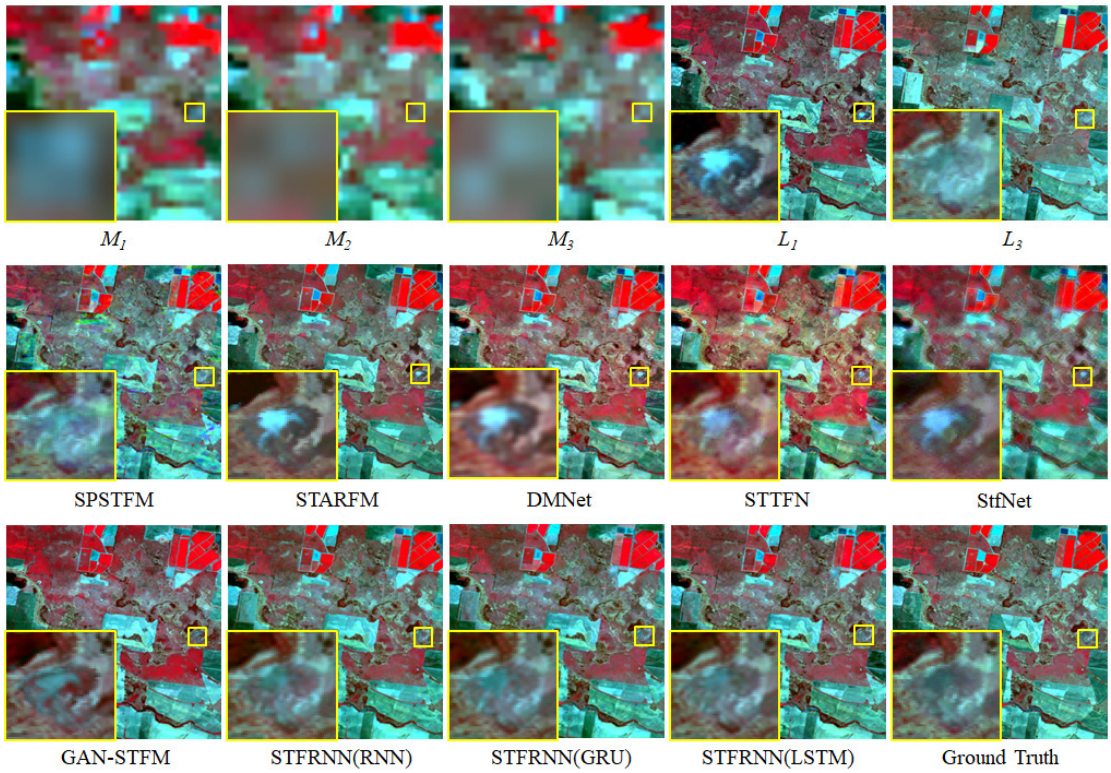



Fig. 10. Fusion results on February 14, 2005, in the LGC dataset. ($M_1$, $M_2$, and $M_3$ are the MODIS images captured on January 29, 2005, February 14, 2005, and March 2, 2005, respectively. $L_1$ and $L_3$ are the Landsat images captured on January 29, 2005, and March 2, 2005, respectively.)

we remove the STSC loss function for comparison, termed $\text{STFRNN}_{\text{noSTSC}}$. The ablation experiments were conducted on the same training dataset and testing dataset with the same network hyperparameters. Tables VI and VII list the quantitative evaluation results on the CIA and LGC datasets, respectively. It is worth mentioning that we list the averages of SSIM and RMSE as an overall indicator. It can be seen that the AWF module can adjust the weight adaptively to improve the accuracy of the fusion result, and the STSC loss function can increase the interpretability of the STFRNN and improve the fusion performance to some degree. In sum, the proposed AWF and STSC loss functions play important roles in the fusion process.

*2) Computational Analysis:* Table X presents the computational cost of all the compared methods, where the complexity is evaluated with the number of floating-point operations per second (FLOPs) and the number of parameters. Since the magnitude of parameters and FLOPs are millions and billions,

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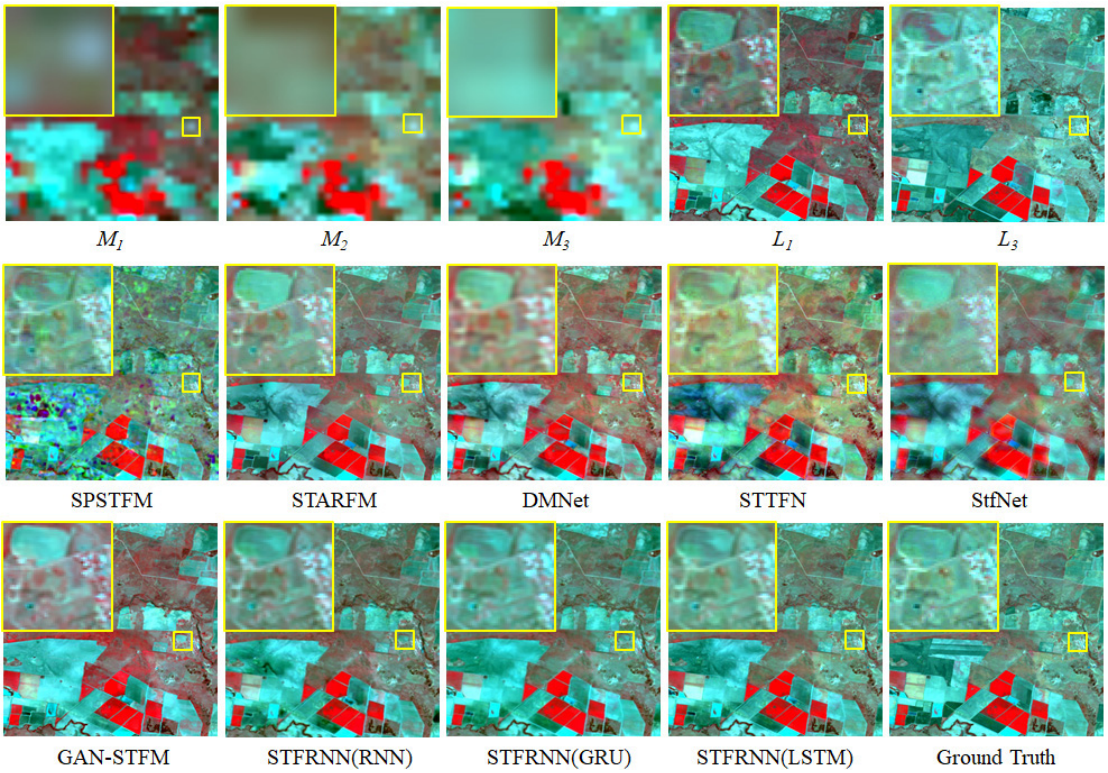



Fig. 11. Fusion results on March 2, 2005, in the LGC dataset. ($M_1$, $M_2$, and $M_3$ are the MODIS images captured on February 14, 2005, March 2, 2005, and April 3, 2005, respectively. $L_1$ and $L_3$ are the Landsat images captured on February 14, 2005 and April 3, 2005, respectively.)

TABLE V

QUANTITATIVE EVALUATION RESULTS IN THE CIA DATASET ON APRIL 27, 2002 (THE BEST IS MARKED IN BOLD, AND THE SECOND BEST IS MARKED IN UNDERLINE)

| | Band | SPSTFM | STARFM | DMNet | StfNet | STTFN | GAN-STFM | STFRNN (RNN) | STFRNN (GRU) | STFRNN (LSTM) |
|---|---|---|---|---|---|---|---|---|---|---|
| RMSE | Band 1 | 0.0102 | 0.0122 | 0.0119 | 0.0155 | 0.0191 | 0.0103 | 0.0113 | 0.0107 | **0.0100** |
| | Band 2 | 0.0149 | 0.0153 | 0.0219 | 0.0183 | 0.0280 | 0.0137 | 0.0147 | 0.0139 | **0.0128** |
| | Band 3 | 0.0186 | 0.0220 | 0.0269 | 0.0274 | 0.0216 | 0.0208 | 0.0240 | 0.0206 | **0.0172** |
| | Band 4 | **0.0305** | 0.0327 | 0.0347 | 0.0396 | 0.0351 | 0.0327 | 0.0408 | 0.0358 | 0.0359 |
| | Band 5 | 0.0352 | 0.0381 | 0.0371 | 0.0418 | 0.0328 | 0.0411 | 0.0411 | 0.0360 | **0.0300** |
| | Band 6 | 0.0296 | 0.0342 | 0.0385 | 0.0361 | 0.0298 | 0.0388 | 0.0335 | 0.0307 | **0.0276** |
| | Average | 0.0232 | 0.0257 | 0.0285 | 0.0298 | 0.0277 | 0.0262 | 0.0276 | 0.0246 | **0.0223** |
| SSIM | Band 1 | 0.9358 | 0.9192 | 0.8541 | 0.8984 | 0.8527 | 0.9387 | 0.9393 | 0.9442 | **0.9485** |
| | Band 2 | 0.9173 | 0.9092 | 0.8599 | 0.9014 | 0.8252 | 0.9215 | 0.9298 | 0.9366 | **0.9406** |
| | Band 3 | 0.8833 | 0.8574 | 0.8218 | 0.8367 | 0.8786 | 0.8680 | 0.8828 | 0.8982 | **0.9088** |
| | Band 4 | 0.8365 | 0.8089 | 0.7995 | 0.8041 | 0.8140 | 0.8159 | 0.8444 | 0.8581 | **0.8666** |
| | Band 5 | 0.7799 | 0.7498 | 0.7882 | 0.7555 | 0.8149 | 0.7593 | 0.7983 | 0.8110 | **0.8298** |
| | Band 6 | 0.7918 | 0.7515 | 0.7706 | 0.7549 | 0.8115 | 0.7628 | 0.8103 | 0.8153 | **0.8303** |
| | Average | 0.8574 | 0.8327 | 0.8157 | 0.8252 | 0.8328 | 0.8444 | 0.8675 | 0.8772 | **0.8874** |
| SAM | All | 0.1467 | 0.1640 | 0.1724 | 0.1711 | 0.1780 | 0.1567 | 0.1501 | 0.1384 | **0.1224** |

we provide MB as the unit for parameters and GB as the unit for FLOPs. Moreover, the testing time on GPU and CPU are also listed, respectively. For these metrics, smaller values indicate better performance. The computational complexity experiments were conducted on the LGC dataset with the size of 512 × 512 × 6. As can be seen, the StfNet and STTFN are with a smaller number of parameters and FLOPs and also spend less testing time due to the simplicity of the network. The DMNet, GAN-STFM, and our STFRNN with a relatively larger number of parameters taking longer testing time due to the complex network. However, both GAN-STFM and STFRNN have a better fusion performance, and our proposed STFRNN achieves the best fusion performance. Moreover, this kind of computational complexity can be basically realized in modern hardware.

*3) Generalization Analysis:* We conducted two experiments to verify the generalization performance of the proposed STFRNN. First, to verify the performance in the case of temporally decoupling, we utilize two pairs of fine-coarse images at $t_1$ (captured on April 16, 2004) and $t_{14}$ (captured

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TABLE VI

QUANTITATIVE EVALUATION RESULTS IN THE LGC DATASET ON FEBRUARY 14, 2005 (THE BEST IS MARKED IN BOLD, AND THE SECOND BEST IS MARKED IN UNDERLINE)

| | Band | SPSTFM | STARFM | DMNet | StfNet | STTFN | GAN-STFM | STFRNN (RNN) | STFRNN (GRU) | STFRNN (LSTM) |
|---|---|---|---|---|---|---|---|---|---|---|
| RMSE | Band 1 | 0.0093 | 0.0093 | 0.0091 | 0.0102 | 0.0098 | 0.0105 | 0.0099 | 0.0086 | **0.0085** |
| | Band 2 | 0.0121 | 0.0113 | 0.0123 | 0.0137 | 0.0115 | 0.0132 | 0.0162 | 0.0101 | **0.0100** |
| | Band 3 | 0.0169 | 0.0147 | 0.0171 | 0.0189 | 0.0164 | 0.0191 | 0.0206 | 0.0147 | **0.0127** |
| | Band 4 | 0.0285 | 0.0276 | 0.0442 | 0.0304 | 0.0341 | 0.0307 | 0.0224 | 0.0240 | **0.0218** |
| | Band 5 | 0.0348 | 0.0354 | 0.0489 | 0.0320 | 0.0255 | 0.0315 | 0.0388 | 0.0208 | **0.0218** |
| | Band 6 | 0.0259 | 0.0288 | 0.0376 | 0.0288 | 0.0242 | 0.0279 | 0.0327 | 0.0192 | **0.0184** |
| | Average | 0.0213 | 0.0212 | 0.0282 | 0.0223 | 0.0203 | 0.0222 | 0.0234 | 0.0162 | **0.0155** |
| SSIM | Band 1 | 0.9686 | 0.9690 | 0.9698 | 0.9607 | 0.9702 | 0.9660 | 0.9605 | 0.9767 | **0.9781** |
| | Band 2 | 0.9591 | 0.9610 | 0.9607 | 0.9465 | 0.9644 | 0.9576 | 0.9591 | 0.9719 | **0.9725** |
| | Band 3 | 0.9347 | 0.9430 | 0.9401 | 0.9114 | 0.9370 | 0.9343 | 0.9467 | 0.9587 | **0.9643** |
| | Band 4 | 0.9034 | 0.9055 | 0.8924 | 0.8624 | 0.8983 | 0.9047 | 0.9303 | 0.9327 | **0.9424** |
| | Band 5 | 0.8779 | 0.8640 | 0.8564 | 0.8401 | 0.8867 | 0.8685 | 0.9072 | 0.9260 | **0.9298** |
| | Band 6 | 0.8967 | 0.8724 | 0.8715 | 0.8641 | 0.8997 | 0.8854 | 0.9133 | 0.9348 | **0.9393** |
| | Average | 0.9234 | 0.9191 | 0.9151 | 0.8975 | 0.9261 | 0.9194 | 0.9362 | 0.9502 | **0.9544** |
| SAM | All | 0.1149 | 0.1110 | 0.1187 | 0.1301 | 0.1104 | 0.1237 | 0.0974 | **0.0851** | 0.0860 |

TABLE VII

QUANTITATIVE EVALUATION RESULTS IN THE LGC DATASET ON MARCH 2, 2005 (THE BEST IS MARKED IN BOLD, AND THE SECOND BEST IS MARKED IN UNDERLINE)

| | Band | SPSTFM | STARFM | DMNet | StfNet | STTFN | GAN-STFM | STFRNN (RNN) | STFRNN (GRU) | STFRNN (LSTM) |
|---|---|---|---|---|---|---|---|---|---|---|
| RMSE | Band 1 | 0.0118 | 0.0109 | 0.0116 | 0.0119 | 0.0117 | 0.0105 | 0.0112 | 0.0107 | **0.0103** |
| | Band 2 | 0.0138 | 0.0133 | 0.0145 | 0.0143 | 0.0129 | 0.0124 | 0.0189 | 0.0126 | **0.0114** |
| | Band 3 | 0.0198 | 0.0199 | 0.0208 | 0.0201 | 0.0168 | 0.0192 | 0.0246 | 0.0173 | **0.0148** |
| | Band 4 | 0.0316 | 0.0244 | 0.0532 | 0.0331 | 0.0305 | 0.0250 | 0.0294 | 0.0258 | **0.0219** |
| | Band 5 | 0.0320 | 0.0329 | 0.0647 | 0.0409 | 0.0358 | 0.0321 | 0.0477 | 0.0262 | **0.0235** |
| | Band 6 | 0.0262 | 0.0408 | 0.0364 | 0.0297 | 0.0243 | 0.0286 | 0.0303 | **0.0184** | 0.0216 |
| | Average | 0.0222 | 0.0237 | 0.0330 | 0.0247 | 0.0220 | 0.0213 | 0.0270 | 0.0185 | **0.0172** |
| SSIM | Band 1 | 0.9676 | 0.9652 | 0.9764 | 0.9686 | 0.9683 | 0.9682 | 0.9521 | 0.9735 | **0.9751** |
| | Band 2 | 0.9536 | 0.9573 | 0.9619 | 0.9517 | 0.9641 | 0.9624 | 0.9565 | 0.9691 | **0.9710** |
| | Band 3 | 0.9298 | 0.9415 | 0.9413 | 0.9198 | 0.9419 | 0.9393 | 0.9422 | 0.9560 | **0.9605** |
| | Band 4 | 0.8945 | 0.9225 | 0.9010 | 0.8846 | 0.9080 | 0.9220 | 0.9365 | 0.9342 | **0.9445** |
| | Band 5 | 0.8861 | 0.8992 | 0.8658 | 0.8516 | 0.8797 | 0.8751 | 0.9106 | 0.9290 | **0.9309** |
| | Band 6 | 0.9016 | 0.9234 | 0.9030 | 0.8917 | 0.9140 | 0.8981 | 0.9241 | **0.9373** | 0.9371 |
| | Average | 0.9222 | 0.9338 | 0.9249 | 0.9113 | 0.9287 | 0.9275 | 0.9370 | 0.9498 | **0.9532** |
| SAM | All | 0.1185 | 0.0934 | 0.0943 | 0.1072 | 0.0990 | 0.1092 | 0.0902 | **0.0857** | 0.0860 |

TABLE VIII

QUANTITATIVE EVALUATION RESULTS IN THE CIA DATASET IN THE ABLATION STUDY (THE BEST IS MARKED IN BOLD)

| Prediction Date | Model | RMSE | SAM | SSIM |
|---|---|---|---|---|
| Apr. 18th 2002 | $\text{STFRNN}_{\text{noAWF}}$ | 0.0236 | 0.1467 | 0.8607 |
| | $\text{STFRNN}_{\text{noSTSC}}$ | 0.0241 | 0.1479 | 0.8599 |
| | STFRNN | **0.0214** | **0.1419** | **0.8679** |
| Apr. 27th 2002 | $\text{STFRNN}_{\text{noAWF}}$ | 0.0242 | 0.1236 | 0.8842 |
| | $\text{STFRNN}_{\text{noSTSC}}$ | 0.0245 | 0.1353 | 0.8747 |
| | STFRNN | **0.0223** | **0.1224** | **0.8874** |

on April 3, 2005) as the auxiliary images to generate the fine image at $t_6$ (captured on October 25, 2004) with the help of coarse image at $t_6$ based on the LGC dataset. The result is listed in Table XI, and we can see that the proposed STFRNN can still achieve competitive performance, especially in the structural similarity. Second, we further train these DL-based methods on the LGC dataset and test them on the CIA dataset to verify the generalization performance of the networks. The results are listed in Table XII, where we can see that the proposed STFRNN can still achieve the best performance.

*4) Advantages and Disadvantages:* As shown in Figs. 6–13 and Tables IV and V, the proposed STFRNN can far improve the fusion performance both in phenology and land cover type changes. The multiscale Siamese CNN can flexibly extract spatio-spectral features of multisource images, and the multilayer convolutional recurrent neural network can fully describe

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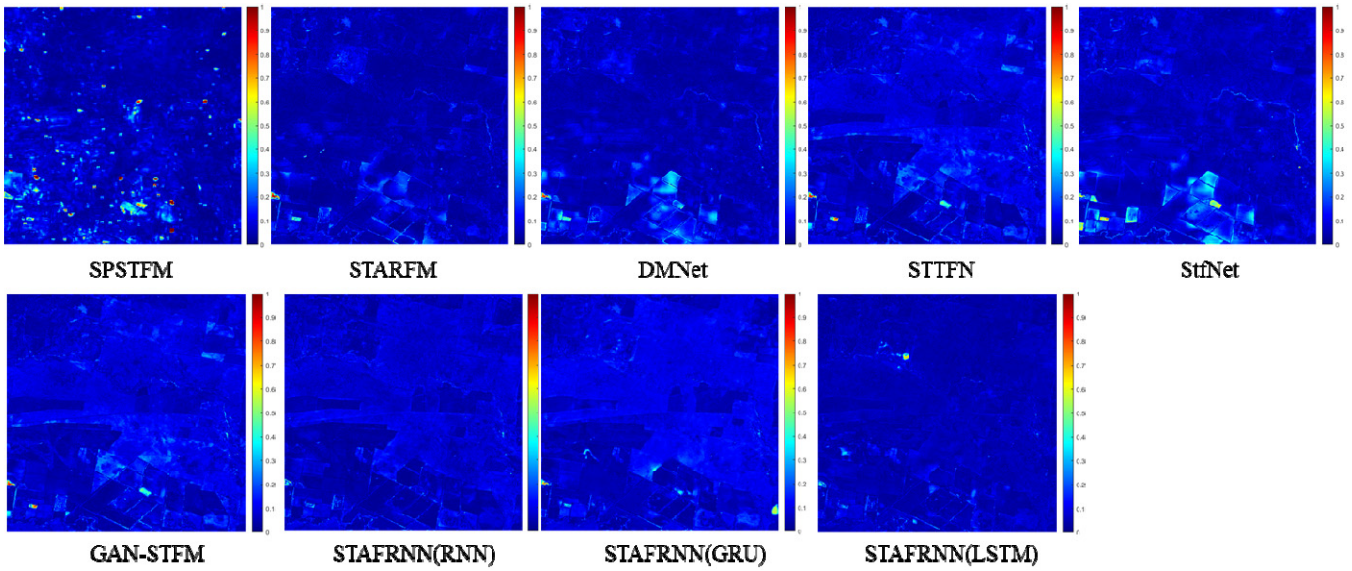



Fig. 12. Heat map of SAM on February 14, 2005, in the LGC dataset.

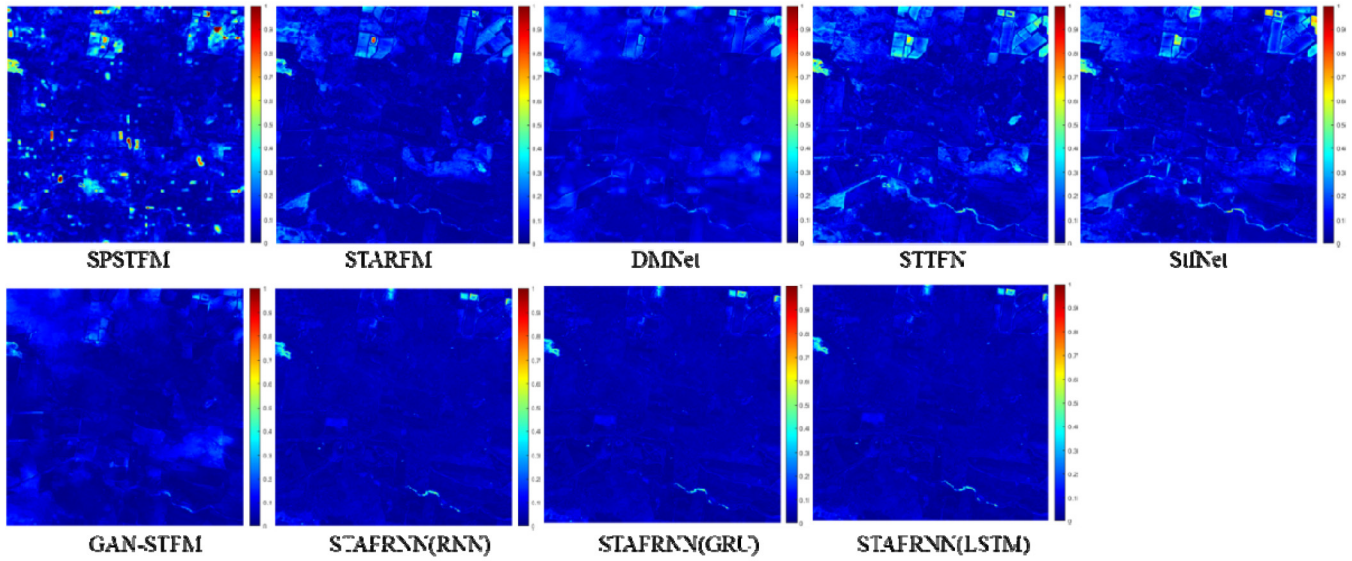



Fig. 13. Heat map of SAM on March 2, 2005, in the LGC dataset.

TABLE IX
QUANTITATIVE EVALUATION RESULTS IN THE LGC DATASET IN THE ABLATION STUDY (THE BEST IS MARKED IN BOLD)

| Prediction Date | Model | RMSE | SAM | SSIM |
|---|---|---|---|---|
| Feb. 14th 2005 | $\text{STFRNN}_{\text{noAWF}}$ | 0.0180 | 0.0905 | 0.9503 |
| | $\text{STFRNN}_{\text{noSTSC}}$ | 0.0181 | 0.0904 | 0.9485 |
| | STFRNN | **0.0155** | **0.0860** | **0.9544** |
| Mar. 2nd 2005 | $\text{STFRNN}_{\text{noAWF}}$ | 0.0203 | 0.0967 | 0.9471 |
| | $\text{STFRNN}_{\text{noSTSC}}$ | 0.0192 | 0.0903 | 0.9474 |
| | STFRNN | **0.0172** | **0.0860** | **0.9532** |

the spatio-temporal relationship of multitemporal images. The AWFNet and reconstruction module synthesizes the spatio-temporal–spectral information to reconstruct the fine image at the target date.

Although STFRNN can effectively improve the fusion performance, it still has some limitations. For example, the model is slightly complicated, which will inevitably increase the training time and test time. To further improve the practicality, we will comprehensively consider the model performance and computational efficiency in our future work.

*5) Quality Evaluation:* In the field of spatio-temporal fusion, the reference image is generally available for a specific

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TABLE X
RESULTS OF PARAMETERS AND EFFICIENCY

| Model | SPSTFM | STARFM | DMNet | STTFN | StfNet | GAN-STFM | STFRNN |
|---|---|---|---|---|---|---|---|
| Parameters (MB) | -- | -- | 0.3271 | 0.0782 | 0.0369 | 0.5832 | 1.2782 |
| FLOPs (GB) | -- | -- | 12.8543 | 1.2807 | 1.2080 | 18.9545 | 16.7650 |
| GPU times(s) | -- | -- | 0.2254 | 0.2014 | 0.1335 | 0.1962 | 0.2016 |
| CPU times(s) | 305 | 366 | 6.2133 | 4.6385 | 1.1642 | 4.1637 | 5.1671 |

TABLE XI
QUANTITATIVE EVALUATION RESULTS TO VERIFY THE GENERATION PERFORMANCE IN THE CASE OF TEMPORAL DECOUPLING IN LGC DATASET (THE BEST IS MARKED IN BOLD, AND THE SECOND BEST IS MARKED IN UNDERLINE)

| | Band | SPSTFM | STARFM | DMNet | StfNet | STTFN | GAN-STFM | STFRNN |
|---|---|---|---|---|---|---|---|---|
| RMSE | Band 1 | 0.0192 | 0.0167 | 0.0137 | 0.0125 | 0.0141 | 0.0143 | **0.0124** |
| | Band 2 | 0.0216 | 0.0176 | 0.0168 | 0.0125 | **0.0117** | 0.0147 | 0.0144 |
| | Band 3 | 0.0269 | 0.0223 | 0.0211 | 0.0168 | 0.0148 | 0.0217 | **0.0135** |
| | Band 4 | 0.0527 | 0.0420 | 0.0444 | 0.0339 | **0.0294** | 0.0431 | 0.0359 |
| | Band 5 | 0.0654 | 0.0483 | 0.0474 | 0.0349 | **0.0323** | 0.0455 | **0.0334** |
| | Band 6 | 0.0764 | 0.0509 | 0.0476 | 0.0444 | **0.0432** | 0.0571 | 0.0449 |
| | Average | 0.0437 | 0.0330 | 0.0318 | 0.0258 | **0.0242** | 0.0327 | 0.0258 |
| SSIM | Band 1 | 0.9065 | 0.9166 | 0.9488 | **0.9652** | 0.9526 | 0.9471 | 0.9568 |
| | Band 2 | 0.9034 | 0.9106 | 0.9372 | 0.9591 | 0.9492 | 0.9459 | **0.9628** |
| | Band 3 | 0.8813 | 0.8838 | 0.9103 | 0.9334 | 0.9202 | 0.9137 | **0.9479** |
| | Band 4 | 0.7337 | 0.7179 | 0.7728 | 0.8074 | 0.7920 | 0.7818 | **0.8481** |
| | Band 5 | 0.7524 | 0.7736 | 0.8048 | 0.8037 | 0.8209 | 0.7965 | **0.8444** |
| | Band 6 | 0.6731 | 0.7339 | 0.7622 | 0.7578 | 0.7727 | 0.7528 | **0.8019** |
| | Average | 0.8084 | 0.8227 | 0.8560 | 0.8696 | 0.8663 | 0.8563 | **0.8936** |
| SAM | All | 0.2217 | 0.1803 | 0.1729 | 0.1450 | 0.1765 | 0.1730 | **0.1338** |

TABLE XII
QUANTITATIVE EVALUATION RESULTS TO VERIFY THE GENERATION PERFORMANCE OF THE SPATIO-TEMPORAL FUSION MODELS ON THE CIA DATASET (ALL DL-BASED MODELS ARE TRAINED ON THE LGC DATASET)

| | Band | SPSTFM | STARFM | DMNet | StfNet | STTFN | GAN-STFM | STFRNN |
|---|---|---|---|---|---|---|---|---|
| RMSE | Band 1 | **0.0102** | 0.0122 | 0.0107 | 0.0118 | 0.0108 | 0.0175 | 0.0109 |
| | Band 2 | 0.0149 | 0.0153 | 0.0163 | 0.0163 | 0.0139 | 0.0195 | **0.0134** |
| | Band 3 | **0.0186** | 0.0220 | 0.0260 | 0.0257 | 0.0215 | 0.0253 | 0.0205 |
| | Band 4 | 0.0305 | 0.0327 | 0.0538 | 0.0390 | 0.0346 | 0.0360 | **0.0272** |
| | Band 5 | 0.0352 | 0.0381 | 0.0672 | 0.0456 | 0.0406 | 0.0452 | **0.0341** |
| | Band 6 | 0.0296 | 0.0342 | 0.0493 | 0.0406 | 0.0341 | 0.0461 | **0.0290** |
| | Average | 0.0232 | 0.0257 | 0.0372 | 0.0298 | 0.0259 | 0.0316 | **0.0225** |
| SSIM | Band 1 | 0.9358 | 0.9192 | 0.9363 | 0.9237 | 0.9395 | 0.8968 | **0.9406** |
| | Band 2 | 0.9173 | 0.9092 | 0.9192 | 0.9095 | 0.9312 | 0.8971 | **0.9353** |
| | Band 3 | 0.8833 | 0.8574 | 0.8642 | 0.8448 | 0.8864 | 0.8432 | **0.8934** |
| | Band 4 | 0.8365 | 0.8089 | 0.8044 | 0.7883 | 0.8394 | 0.7988 | **0.8691** |
| | Band 5 | 0.7799 | 0.7498 | 0.7459 | 0.7359 | 0.7857 | 0.7167 | **0.8170** |
| | Band 6 | 0.7918 | 0.7515 | 0.7560 | 0.7550 | 0.8077 | 0.7160 | **0.8222** |
| | Average | 0.8574 | 0.8327 | 0.8377 | 0.8262 | 0.8650 | 0.8114 | **0.8796** |
| SAM | All | 0.1467 | 0.1640 | 0.1571 | 0.1679 | 0.1436 | 0.1836 | **0.1401** |

target phase, thus the quantitative evaluation of the fused image is commonly founded on the full-reference quality evaluation indexes. However, a nonreference quality evaluation metric is required to evaluate the fused image if the reference image is unavailable due to cloud coverage or other factors. In addition, for the quality evaluation of the spatio-temporal fusion, the temporal consistency should be considered for time-series high spatial resolution fused images. In this case, the reference images are generally unavailable, and a blind quantitative evaluation index should be developed by considering the temporal consistency, spectral fidelity, and spatial sharpness of the fused images.

## IV. CONCLUSION

In this article, we propose a spatio-temporal–spectral collaborative learning framework for the spatio-temporal fusion of remote-sensing images with land cover changes. In the proposed framework, the multiscale Siamese CNN is designed to extract effective features of multisource and multitemporal

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remote-sensing images captured by different sensors. The multilayer convolutional recurrent neural network with ConvLSTM is constructed to comprehensively extract the land cover changes on spatial, spectral, and temporal joint features. In addition, the AWFNet and the STSC loss are proposed to improve the fusion performance. The quantitative and qualitative experiments are conducted on the opening benchmark datasets with changes in phenology and land cover, respectively. The experimental results demonstrated that the proposed STFRNN achieves competitive performance.

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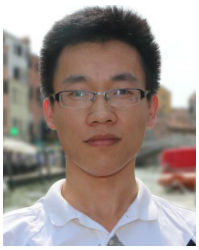
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